

Inert Escape-room Console

Solve six clue stations without opening anything

E0 POLICY COMPOSITION — NO DOOR, LOCK, LATCH, LAMP, OR POWER PATH

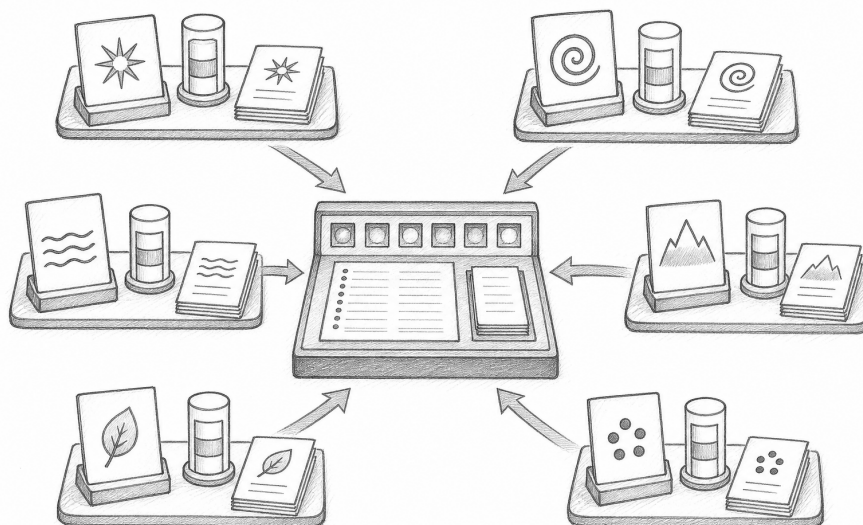
Lesson 057 — project
Prerequisites Lessons 055 and 056
Evidence named memory result cells

Time 180–240 minutes
Board compile-only Mega replay
Boundary copied values and inert intent

Project boundary. `InertEscapeConsole` owns a fixed clue model and a fault-aware operator panel. It publishes only copied lamp, presentation, and demonstration-latch intent. It owns no endpoint, pin, interrupt, timer, display, storage medium, servo, relay, lamp, latch, door, lock, or power path. It is not access control, confinement, egress, an emergency release, a protective interlock, or a life-safety system.

1 Tour the six fictional stations

Twelve copied clues form six fixed pairs. The family mapping is part of the configuration digest; runtime input cannot rename or rearrange it.



Alternative text: a graphite tabletop console is surrounded by six clearly separate fictional stations, each with two symbolic clue cards. Arrows carry their copied results to a central policy notebook. The adjacent table gives the exact Sequence, Pattern, Orientation, Presence, Rhythm, and Alignment mapping; no wiring, door, lock, or powered actuator is present.

Family	Clues	What the E0 replay observes
Sequence	0–1	two copied categorical values and their provenance
Pattern	2–3	two independent evidence cells, never an inferred sensor
Orientation	4–5	symbolic categories, not a physical pose measurement
Presence	6–7	synthetic presence categories, not occupancy or identity
Rhythm	8–9	copied categories, not audio sampling or timing hardware
Alignment	10–11	synthetic alignment, not position or motion authority

Every family summary reports its clue IDs, completeness, and weakest quality. It cannot grant solve authority. Only the complete twelve-rule DAG, qualified stop evidence, panel state, audit state, and a deliberate `prepareSolve()` can do that.

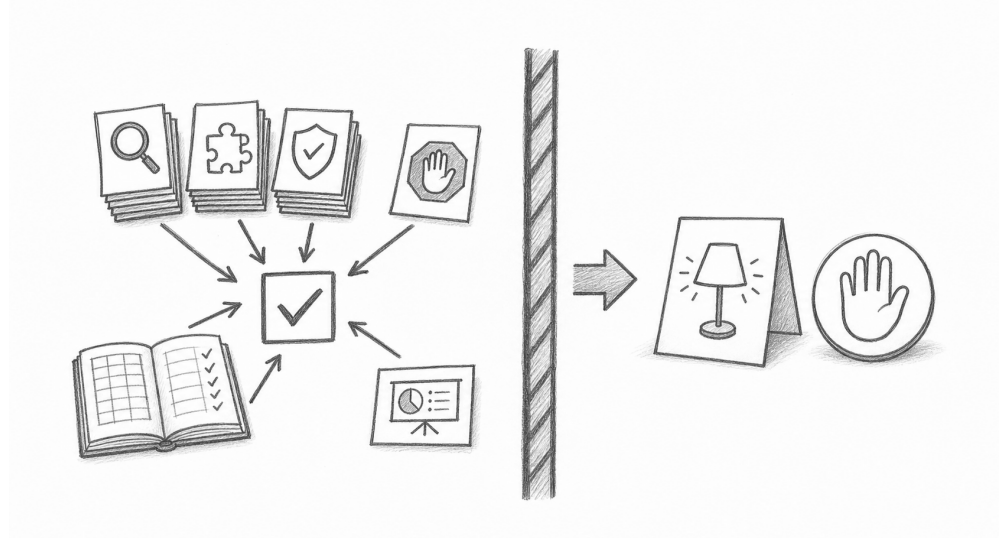
2 Submit one complete atomic envelope

`EscapeConsoleUpdate` carries one supplied time and presence-qualified copies of the audit image, clue update, stop, control, audit acknowledgement, diagnostic acknowledgement, presentation evidence, and solve preview. Noncanonical absent values, malformed evidence, or a structurally invalid image reject the whole envelope without mutation.

After structural validation, precedence is fixed:

1. retained lifecycle, configuration, source, and timing faults;
2. qualified stop assertion or an unreconciled stop transition;
3. clue faults, stale evidence, contradiction, and invalid chords;
4. corrupt or indeterminate audit state and presentation failure;
5. ordinary panel audit or diagnostic acknowledgement;
6. exact solve commit; and
7. progress, confirmation, solved, or inactive presentation intent.

There is no public clue, stop, acknowledge, solve, show, latch, or lamp call whose caller-selected order can change that precedence.



Alternative text: a hand-drawn envelope contains clue, stop, control, presentation, audit-image, acknowledgement, and solve-preview cards. A graphite checklist validates the complete envelope before two child notebooks and one parent result cell change together. A bold stop card diverts the envelope to inactive latch and stopped lamp intent before the solve path.

3 Read predictions from named memory cells

Cell	Prediction	Interpretation
six family cells	incomplete, then complete with weakest quality	copied clue grouping only
constraint cell	exact clue generation and satisfied-rule mask	fixed DAG result, not physical truth
panel cell	diagnostic, selected station, stop and audit disposition	copied operator policy only
audit image	latest two canonical records	modeled memory, not persistence
lamp cell	off, ready, progress, confirmation, solved, fault, or stopped	semantic presentation intent only
latch cell	inactive except one solved transaction	request for an absent demonstration actuator

These volatile cells are the E0 non-Serial observation path. Compilation does not prove that a powered Mega executed the sketch or that any physical indicator displayed the result.

4 Prepare and bind one solution

`prepareSolve(operationId, now)` is the deliberate operator confirmation. A selected panel cell, a Select control, or twelve satisfied rules cannot silently substitute for this named call.

Preparation requires a solved retained clue generation, all six families complete, fresh qualified deasserted stop evidence, no pending stop audit, clear dominant diagnostics, and a canonical audit image that is **Ready** or the exact empty-image **PrepareRequired** bootstrap. It privately prepares the child **PuzzleSolved** record.

The returned preview binds:

- parent owner token, lifecycle, configuration revision, and epoch;
- console generation, operation ID, policy digest, clue generation, and satisfied-rule mask; and
- the complete child audit preview, including its owner, lifecycle, sequence, slot, record, image digest, and predecessor.

`canCommit()` answers only whether those retained preview fields are still live. It does not validate a later image or promise that a later `update()` will commit. A foreign, stale, changed, consumed, reset-invalidated, child-mismatched, or image-mismatched preview rejects.

5 Commit parent and child together

The caller models a write by copying the prepared child record into the specified audit-image slot. A later complete envelope supplies that image, the exact child acknowledgement nested in the parent preview, and the exact parent preview.

The coordinator first prepares the complete proposed clue and panel states. It recomputes families, diagnostics, presentation, latch intent, lamp intent, policy identity, and every preview binding from those proposed values. Only after every parent and child preflight succeeds do the infallible apply seams mutate both children and publish one parent transition.

Collision in the same envelope	Required result
stop assertion plus solve	stop wins; neither capability is consumed
clue becomes stale or contradictory plus solve	solve is suppressed
presentation fails plus solve	fault intent; no solved audit commit
invalid chord plus solve	invalid chord; inactive latch intent
wrong or torn audit image plus solve	reject or fail closed; no partial state
exact canonical image and both previews	one Solved transition

Sequential best-effort mutation and rollback after an observable child change are outside the design. If implementation ever needs either, stop promotion and repair the architecture.

6 Make stop and restart boring

A default **stopped == false** value is not qualified evidence. Missing, stale, faulted, or asserted stop evidence rejects solution preparation. A qualified asserted observation publishes stopped intent and requires its exact **StopAsserted** audit record. A later qualified deasserted observation still leaves latch intent inactive until the exact newer **StopReleased** record commits.

Stage	Predict	Observe
stop collides with solve	stopped dominates	inactive latch; no solved record
asserted record commits	stopped remains retained	canonical audit sequence
deassertion arrives	release still pending	inactive latch; pending audit
released record commits	ordinary puzzle policy resumes	canonical newer record
shutdown and initialize	blank volatile state	inactive/off; no image assumed
clean image replay	deterministic reconciliation	Ready or retained Stopped
torn image replay	fail closed	Corrupt or Indeterminate; inactive latch

Software stop is neither emergency stop nor power removal. Even a later physical demonstration needs an independent means to remove load power.

7 Run the deterministic console workbook

1. Initialize without an image. Predict inactive latch intent, lamp off, six incomplete families, and an unreconciled audit disposition.
2. Supply every one of the 2^{12} clue-presence masks. Confirm that each family becomes complete only when both mapped clues are eligible.
3. Replay the twelve-rule chain from incomplete through solved. Inject stale, contradictory, source-faulted, wrong-source, future, regressed, and half-range-ambiguous evidence.
4. Exercise all 16 low-nibble control masks. Confirm zero means no control, one bit means one control, and every multi-bit chord is invalid.
5. Fail presentation, recover it with exact current-generation evidence, prepare its acknowledgement, and commit the attributed audit record.
6. Prepare a solve from the empty-image bootstrap and from a normal Ready image. Mutate each parent and child preview field separately.
7. Collide solve with stop, clue changes, presentation failure, invalid chord, audit ambiguity, and mismatched images. Prove no partial commit.
8. Complete stop assertion and release audits, then restart from clean, torn, corrupt, indeterminate, and adjacent committed images.
9. Replay two configured instances field-for-field and compare every snapshot and canonical audit field.

stage	families	rule mask	panel	audit	latch/lamp
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____

8 Separate acquisition from safe state

Acquisition prediction and observation. Canonical fixed configuration, nonzero parent and child revisions and epochs, twelve source identities, six exact family pairs, and a matching derived policy digest allow initialization. Host tests inspect lifecycle generations, object bindings, clue and panel snapshots, canonical audit fields, and distinct result cells.

Acquisition interpretation. This proves fixed-storage E0 policy construction only. It does not acquire a sensor, control, stop circuit, display, storage medium, lamp, latch, actuator, or power path.

Safe-state prediction and observation. Construction, failed initialization, reset, shutdown, stop, fault, invalid evidence, audit ambiguity, and restart begin or remain at inactive latch intent. Host tests observe that copied state directly.

Safe-state interpretation. Inactive memory intent is not electrical isolation, an emergency stop, a released exit, or proof of a physical latch, lamp, or door state.

9 Read the measured resource evidence

The exact no-LTO maximum-composition Mega 2560 probe measures 34,978 bytes of flash and 3,655 bytes of static SRAM. `sizeof(InertEscapeConsole)` is exactly 1,024 bytes. Conservative compiler evidence measures a 951-byte deepest synchronous stack path. After adding a separate 128-byte interrupt reserve, the residual SRAM is $8192 - 3655 - 951 - 128 = 3458$ bytes.

The object, stack, static-SRAM, and residual values pass their targets. Flash misses the 32,768-byte target but passes the reviewed 40,960-byte hard limit; the fixed-capacity value-owned children and atomic parent/child path are retained rather than weakening ownership or solve semantics. The checked-in `escape-console-resource-check` binds the accepted target miss to this exact measured tuple and review record.

These are compiler, linker, and conservative accounting results for exact fixtures and a recorded toolchain. They are not runtime high-water measurements, physical timing evidence, or proof of any hardware endpoint. Rerun the probe after any source, toolchain, optimization, library, fixture, buffer, or aggregate-composition change.

10 Keep powered work outside E0

E1 must separately qualify the exact passive clue and control inputs, low-voltage current-limited presentation hardware, complete pin and electrical data, pull policy, source identity, named observation points, unpowered wiring inspection, failure injection, and bench acceptance. Servo, relay, latch, lock, door, and other powered actuation remain physically absent.

E2 may consider only a restrained no-load demonstration servo or an inert current-limited low-voltage relay-and-lamp fixture, with independent physical load-power removal and recorded bench acceptance. It must never attach to a door, gate, lock, occupied enclosure, alarm, emergency lighting, egress route, or life-safety system. No person, animal, property, or safe exit may depend on this project.

11 Record the E0 result

- policy digest: _____ final clue generation: _____
- satisfied-rule mask: _____ final audit sequence: _____
- solved transaction: _____ stop-dominance result: _____
- restart result: _____ inactive safe-state result: _____